Carlos Alberto Rios Naranjo

Frontend Architect | Angular · React · Next.js · NestJS

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PROFESSIONAL SUMMARY

Frontend Architect with 5+ years designing, scaling and modernizing enterprise web platforms in Angular, React and Next.js. Defined a multi-brand, multi-site microfrontend architecture (Module Federation, single-spa, headless CMS), delivered platforms serving 100K+ users and 17,000+ suppliers, cut build times by 25% and improved Core Web Vitals by 35%. Currently modernizing production Angular applications from Angular 18 to 22, adopting standalone components, Signals and Native Federation. As founding engineer at a sustainable-finance startup, also built a Next.js + NestJS platform end to end, bringing a full-stack, product-minded view to frontend architecture decisions.

TECHNICAL SKILLS

- **Angular:** Angular (12–22), Signals, standalone components, NgRx, RxJS, Angular Material, PrimeNG
- **React:** React, Next.js, Redux Toolkit, React Query, React Native (Expo)
- **Languages & styling:** TypeScript, JavaScript, HTML, CSS, Tailwind CSS
- **Architecture:** Microfrontends (Module Federation, Native Federation, single-spa), Nx, Turborepo, Headless CMS (Umbraco, Directus), Storybook
- **Testing:** Jasmine, Karma, Jest, Vitest, React Testing Library, Playwright (E2E)
- **Backend & data:** NestJS, PostgreSQL, TypeORM, REST APIs, ERP integrations (QuickBooks, Alegra)
- **DevOps & cloud:** Docker, GitHub Actions, AWS, Hetzner Cloud
- **Practices:** Framework upgrades and migrations, performance optimization (Core Web Vitals), code reviews, technical documentation, Scrum

PROFESSIONAL EXPERIENCE

Frontend Developer | Apex Global Mobility | Nov 2025 – Present

- Upgraded 24 internal Angular repositories across major versions (18 → 20, now moving to 22) as part of the frontend team, resolving breaking changes and keeping dependencies current.
- Migrate internal projects to standalone components and Signals-based reactivity, simplifying module structure and state handling.
- Deliver new features end to end in a microfrontend architecture based on Native Federation, with peer code reviews on every pull request.

Founding Engineer (part-time) | Cosmo (early-stage sustainable-finance startup) | Jun 2025 – Oct 2026

- Built a three-sided B2B sustainable-finance platform connecting companies, banks and investors: GHG accounting, EU/LATAM taxonomy, IFRS S1/S2 reporting, green-credit offers and ESG portfolios.
- Architected a TypeScript monorepo (Turborepo, Yarn workspaces) with a Next.js 16 web app, a NestJS 11 API, TypeORM/PostgreSQL and a shared Storybook design system.
- Integrated QuickBooks Online and Alegra ERPs with OAuth token refresh, scheduled incremental syncs and tested data transforms feeding the expense and revenue modules.
- Set up CI/CD with GitHub Actions (Jest, Vitest, Playwright E2E, Trivy security scans) and Docker deployments on Hetzner Cloud with AWS S3 backups.

Frontend Architect | Inchcape Digital | Jun 2024 – Feb 2025

- Designed a multi-brand, multi-site, multi-language frontend architecture with Next.js, Module Federation and single-spa, integrated with Umbraco as headless CMS.
- Defined secure cookie sharing across subdomains to deliver unified authentication and consistent sessions between brand sites.
- Built shared UI components and moved builds to a Turborepo monorepo, cutting build times by 25%.

- Established documentation standards in Confluence and ran onboarding workshops for new teams.
- Led code reviews and performance practices with backend, UX and product teams.

Frontend Developer (Development Analyst) | XpertGroup S.A.S. | Jun 2022 – Jun 2024

- Developed Puntos Colombia for Businesses, a platform serving 100K+ users, with Angular.
- Implemented NgRx and RxJS state management for complex business workflows, improving responsiveness and maintainability.
- Built a reusable component library with Angular Material and Storybook.
- Contributed to frontend architecture definitions, unit testing with Jasmine and Karma, code reviews and Docker-based CI.

Frontend Developer | Intelcost S.A.S. (Suplos) | Jun 2021 – Jun 2022

- Rebuilt the UI of Intelcost's supply-chain platform, serving 17,000+ suppliers, with Angular, TypeScript and RxJS.
- Improved Core Web Vitals by 35% through lazy loading and bundle splitting.
- Integrated real-time REST APIs with RxJS data streams, covered by Jasmine and Karma tests.

Freelance Full Stack Developer | Jul 2019 – May 2021

- Designed and delivered end-to-end web and mobile solutions for clients, from architecture to deployment.
- Built with React, Angular, React Native (Expo), Node.js, NestJS, PostgreSQL, MySQL and Directus.
- Automated deployments with Docker and GitHub Actions; advised clients on product architecture and MVP scope.

PERSONAL PROJECTS

Cross-chain NFT launch — Blockchain Bites bootcamp capstone | 2023 | github.com/CarlosAlbertoR/batch-02-desafio-integrador

- Implemented upgradeable (UUPS) ERC-20 and ERC-721 contracts in Solidity with OpenZeppelin, role-based access, Merkle-tree whitelist minting and a burn-for-buy-back flow, covered by 32 Hardhat unit tests.
- Built a public-sale contract accepting tokens, USDC (swapped through Uniswap V2) or ether, deployed to the Ethereum Goerli and Polygon Mumbai testnets.
- Relayed events between chains with OpenZeppelin Defender Autotasks and Sentinels; hosted NFT metadata on IPFS and built the Ethers.js frontend.

EDUCATION

Systems and Telecommunications Engineer | University of Manizales | Jun 2023

Thesis: Measurement of vehicle combustion gas concentration in Manizales using UAVs and GIS — Arduino sensors mounted on a drone to sample at several altitudes, with GIS analysis of the spatial distribution of urban air pollution.

CERTIFICATIONS

McKinsey.org Forward Program — structured problem-solving, communication, adaptability, digital and AI essentials | McKinsey.org | Dec 2025 | Credential

Scrum Foundation Professional Certificate (SFPC) | CertiProf | Nov 2022

LANGUAGES

Spanish (native) · English (B2)